

CIE IGCSE COMPUTER SCIENCE (0478)

Programme Support Document

5-MOCK EXAM SERIES & GAP ANALYSIS SYSTEM

五轮模拟考试系列与差距分析系统

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1. OVERVIEW | 概述

The 5-Mock Exam Series is a structured assessment programme designed to prepare students systematically for the CIE IGCSE Computer Science (0478) examinations.

Each mock exam targets specific skills, identifies knowledge gaps, and informs targeted teaching interventions.

五轮模拟考试系列是一项结构化评估方案，旨在系统地为学生备考剑桥 IGCSE 计算机科学 (0478) 考试做好准备。每次模拟考试针对特定技能，识别知识差距，并为有针对性的教学干预提供依据。

Key Principles | 核心原则:

- Progressive skill building — theory first, then programming
- Data-driven instruction — every mock feeds into teaching decisions
- Student ownership — learners track their own progress
- Early identification — gaps found early, closed quickly

2. MOCK EXAM CALENDAR | 模拟考试日历

The five mock examinations are distributed across Year 10 to maximise preparation time while allowing sufficient gaps for targeted revision between assessments.

五场模拟考试分布在九年级全年，以最大化备考时间，同时在每次评估之间留出足够的针对性复习间隔。

MOCK EXAM CALENDAR — YEAR 10 → 九年级模拟考试日历

Mock → Date → Paper → Purpose & Focus

编号 → 日期 → 试卷 → 目的与重点

Mock 1 → Y10S1 W17 → Paper 1 → Baseline assessment. Full theory

模拟 1 → 第一学期第 17 周 → (75 marks) → paper covering all Units 1–8 topics

试卷一 → taught so far. Establishes starting

(75 分) → point for every student.

基线评估。涵盖已授单元 1-8 全部理论

考卷。确立每位学生的起点。

Mock 2 → Y10S2 W9 → Paper 1 → Progress check. Same format as

模拟 2 → 第二学期第 9 周 → (75 marks) → Mock 1. Measures knowledge retention

试卷一 → and identifies persistent weak areas.

(75 分)

进度检查。与模拟 1 格式相同。衡量知识

保持度并识别持续薄弱领域。

Mock 3 → Y10S2 W10 → Paper 2 → Programming assessment. Full Paper 2

模拟 3 → 第二学期第 10 周 → (75 marks) → covering problem-solving, algorithms,

试卷二 → pseudocode, and flowcharts.

(75 分)

编程评估。完整试卷二，涵盖问题解决、

算法、伪代码和流程图。

Mock 4 → Y10S2 W15 → Paper 1 → Pre-exam drill. Full Paper 1 under

模拟 4 → 第二学期第 15 周 → (75 marks) → exam conditions. Focus on exam

试卷一 → technique and time management.

(75 分)

考前演练。完整试卷一，考试条件下进行。

重点训练考试技巧和时间管理。

Mock 5 → Y10S2 W16 → Paper 2 → Final programming drill. Full Paper 2

模拟 5 → 第二学期第 16 周 → (75 marks) → under exam conditions. Final gap

试卷二 → analysis before external exams.

(75 分)

最终编程演练。完整试卷二，考试条件下。

外部考试前的最终差距分析。

Note: Y10S1 = Year 10 Semester 1 | Y10S2 = Year 10 Semester 2

W17 = Week 17, etc.

3. PURPOSE OF EACH MOCK | 每次模拟的目的

MOCK 1 — Diagnostic Baseline | 诊断基线

- Tests: All theory topics from Units 1–8 covered in Semester 1
- When: End of Semester 1, before holiday break
- Results used to:
 - Identify individual student starting points
 - Form initial revision groups (by topic weakness)
 - Adjust Semester 2 teaching priorities
 - Set personalised targets for each student
- 测试：第一学期已授单元 1–8 全部理论主题
- 结果用于：识别个人起点、组建复习小组、调整教学重点、设定个性化目标

MOCK 2 — Progress Measurement | 进度测量

- Tests: Theory knowledge retention after holiday break
- When: Early Semester 2, after targeted revision period
- Results used to:
 - Compare directly with Mock 1 to show improvement
 - Confirm whether gap-closing strategies worked
 - Identify topics that need continued attention
 - Refine revision groupings
- 测试：假期后的理论知识保持度
- 结果用于：与模拟 1 对比显示进步、确认差距缩小策略是否有效

MOCK 3 — Programming Skills | 编程技能

- Tests: Paper 2 skills — algorithms, pseudocode, logic, problem-solving
- When: Immediately after Mock 2, while theory is fresh
- Results used to:
 - Assess programming readiness for external exams
 - Identify students needing intensive coding practice
 - Balance theory vs. programming revision time
 - Flag students for one-to-one programming support
- 测试：试卷二技能——算法、伪代码、逻辑、问题解决
- 结果用于：评估编程备考状态、识别需加强编程练习的学生

MOCK 4 — Theory Exam Readiness | 理论考试准备度

- Tests: Full Paper 1 under strict exam conditions
- When: Mid-late Semester 2, after all content taught
- Results used to:
 - Fine-tune exam technique (timing, question approach)
 - Identify last-minute theory gaps
 - Prioritise final revision topics
 - Build student confidence through rehearsal
- 测试：严格考试条件下的完整试卷一
- 结果用于：微调考试技巧、识别最后的理论差距、建立学生信心

MOCK 5 — Programming Exam Readiness | 编程考试准备度

- Tests: Full Paper 2 under strict exam conditions
- When: Final mock before external exams
- Results used to:
 - Confirm programming exam readiness
 - Complete final gap analysis
 - Allocate last support resources effectively
 - Provide students with clear action plan for independent study
- 测试：严格考试条件下的完整试卷二
- 结果用于：确认编程考试准备度、完成最终差距分析

4. GAP ANALYSIS PROCESS | 差距分析流程

After each mock exam, a four-step gap analysis process is conducted:

每次模拟考试后，执行四步差距分析流程：

STEP 1: INDIVIDUAL GAP IDENTIFICATION

第一步：个人差距识别

Each student receives their marked paper and completes a self-analysis:

每位学生收到批改后的考卷并完成自我分析：

1. Review every incorrect answer

检查每道错题

2. Classify errors by topic (e.g., "Binary", "Networks", "Logic Gates")

按主题分类错误（如“二进制”“网络”“逻辑门”）

3. Identify your TOP 3 weakest topics

识别你的三大最薄弱主题

4. Write one specific action for each weak topic

为每个薄弱主题写出一个具体行动

Example student entry:

Weak Topic 薄弱主题 → Marks Lost 失分 → Action Plan 行动计划

Binary & Hexadecimal → 8 → Complete extra worksheet set 3A

二进制与十六进制 → and watch video tutorial #12

完成额外练习册 3A，观看视频教程#12

Network Security → 6 → Review Unit 5 notes pages 24–31

网络安全 → and attempt past paper Qs

复习单元 5 笔记 24-31 页，练习历年真题

Logic Gates → 5 → Use online logic gate simulator

逻辑门 → for 30 minutes daily

每日使用在线逻辑门模拟器 30 分钟

STEP 2: CLASS-LEVEL GAP ANALYSIS

第二步：班级层面差距分析

The teacher collates all individual analyses to identify class-wide patterns:

教师汇总所有个人分析，识别全班性模式：

1. Which topics did MOST students struggle with?

大多数学生在哪些主题上遇到困难？

2. Which topics had the lowest class average?

哪些主题的班级平均分最低？

3. Which skills (recall, application, evaluation) were weakest?

哪些技能（回忆、应用、评估）最薄弱？

Class-level data drives:

- Whole-class revision lessons

全班复习课

- Re-teaching of specific concepts

特定概念的重新教学

- Adjustment of remaining lesson time allocation

调整剩余课时分配

STEP 3: TARGETED REVISION GROUPS

第三步：针对性复习小组

Students are grouped by shared weak areas for focused revision sessions:

学生按共同薄弱领域分组，进行集中复习：

Grouping Method 分组方法:

- Group A: Binary / Hex / Denary conversion struggles

A 组：二进制/十六进制/十进制转换困难

- Group B: Network concepts and protocols

B 组：网络概念与协议

- Group C: Programming logic and algorithms

C 组：编程逻辑与算法

- Group D: Data representation and storage

D 组：数据表示与存储

- Group E: Security, ethics, and emerging technologies

E 组：安全、伦理与新兴技术

Session Format session 格式:

- 20–30 minutes per session

每次 20–30 分钟

- Differentiated worksheet or activity for each group

每组差异化练习或活动

- Peer teaching component — students help each other

同伴教学环节——学生互助

- Teacher rotates between groups for support

教师轮流支持各组

STEP 4: PROGRESS TRACKING

第四步：进度追踪

Every student's mock results are tracked across all five exams to show:

每位学生的模拟成绩在五场考试中全程追踪，以显示：

1. Overall trend (improving / stable / declining)

总体趋势（improving 改善 / stable 稳定 / declining 下降）

2. Topic-by-topic improvement

逐主题改善情况

3. Readiness for external exams

外部考试准备度

See Sample Progress Tracker below.

见下方样本进度追踪表。

5. SAMPLE GAP ANALYSIS TEMPLATE (BLANK) | 样本差距分析模板 (空白)

STUDENT GAP ANALYSIS FORM 学生差距分析表

Student Name 姓名: _____ Class 班级: _____

Mock Exam 模拟考试: Mock 1 [] Mock 2 [] Mock 3 [] Mock 4 [] Mock 5 []

Date 日期: _____

Topic 主题 → Score/Max → Action Plan 行动计划

得分/满分

Topic 1: _____ → ____/____ → 1. _____

2. _____

Topic 2: _____ → ____/____ → 1. _____

2. _____

Topic 3: _____ → ____/____ → 1. _____

2. _____

Topic 4: _____ → ____/____ → 1. _____

2. _____

Topic 5: _____ → ____/____ → 1. _____

2. _____

TOP 3 PRIORITY TOPICS 三大优先主题:

1. _____ 2. _____

3. _____

Target Score for Next Mock 下次模拟目标分数: _____ / 75

Student Signature 学生签名: _____ Date 日期: _____

Teacher Signature 教师签名: _____ Date 日期: _____

6. SAMPLE PROGRESS TRACKER | 样本进度追踪表

INDIVIDUAL STUDENT PROGRESS TRACKER 个人学生进度追踪表

Student Name 学生姓名: _____

Target Grade 目标等级: _____ CIE Grade Boundary 剑桥等级分界线: A*=68

Mock 1 → Mock 2 → Mock 3 → Mock 4 → Mock 5

模拟 1 → 模拟 2 → 模拟 3 → 模拟 4 → 模拟 5

Y10S1 → Y10S2 → Y10S2 → Y10S2 → Y10S2

W17 → W9 → W10 → W15 → W16

Paper 1 Score → ____/75 → ____/75 → — → ____/75 → —

试卷一得分

Paper 2 Score → — → — → ____/75 → — → ____/75

试卷二得分

% Score → % → % → % → % → %

百分比得分

Projected Grade → /A-E → /A-E → /A-E → /A-E → /A-E

预测等级

Gap 1 (Top Weak) → _____ → _____ → _____ → _____ → _____

差距 1 (最薄弱)

Gap 2 → _____ → _____ → _____ → _____ → _____

差距 2

Gap 3 → _____ → _____ → _____ → _____ → _____

差距 3

Intervention → _____ → _____ → _____ → _____ → _____

Received 已接受干预

On Track? 进度正常? → Yes/No → Yes/No → Yes/No → Yes/No → Yes/No

CIE Grade Boundaries (approximate) 剑桥等级分界线 (近似值):

A* = 68-75 | A = 60-67 | B = 52-59 | C = 44-51 | D = 36-43 | E = 28-35

7. HOW DATA DRIVES TEACHING DECISIONS | 数据如何驱动教学决策

Mock exam data is used systematically to inform instructional choices:

模拟考试数据被系统化地用于指导教学选择:

1. Curriculum Pacing | 课程节奏

- If >50% of class score below 50% on a topic → re-teach before advancing
若超过 50% 的学生在某主题得分低于 50% → 在 advance 之前重新教学
- Class average >70% on a topic → reduce time, move to enrichment
班级平均分高于 70% → 减少时间, 转向拓展内容

2. Differentiation | 差异化教学

- Students scoring <40% → intensive support group + modified materials

得分低于 40% 的学生 → 强化支持小组 + 调整材料

- Students scoring 40–65% → targeted revision group + gap worksheets

得分 40–65% 的学生 → 针对性复习小组 + 差距练习

- Students scoring >65% → extension tasks + peer tutoring roles

得分高于 65% 的学生 → 拓展任务 + 同伴辅导角色

3. Resource Allocation | 资源分配

- Identify which worksheets, videos, and activities are most needed

识别最需要哪些练习册、视频和活动

- Prioritise creation of resources for repeatedly weak topics

优先为持续薄弱的主题创建资源

- Deploy teaching assistant time to highest-need areas

将教学助理时间部署到最需要帮助的区域

4. Parent Communication | 家长沟通

- Mock results shared with parents after each assessment

每次评估后与家长分享模拟成绩

- Specific action plans provided for home support

为家庭支持提供具体行动计划

- Progress tracker updated and shared each semester

每学期更新并分享进度追踪表

5. Intervention Triggers | 干预触发条件

- Mock score drops by >10 marks → immediate one-to-one meeting

模拟分数下降超过 10 分 → 立即进行一对一会谈

- Same topic weak across 2+ mocks → intensive intervention block

同一主题在 2 次以上模拟中薄弱 → 强化干预模块

- Student below target grade boundary → parent meeting + action plan

学生低于目标等级线 → 家长会议 + 行动计划

8. SUMMARY | 总结

The 5-Mock Exam Series & Gap Analysis System ensures that:

五轮模拟考试系列与差距分析系统确保：

- ✓ No student falls behind without early identification

没有学生会在未被及早发现的情况下落后

- ✓ Teaching is continuously adjusted based on real data

教学会根据真实数据持续调整

✓ Students take ownership of their own learning journey

学生对自己的学习旅程负起责任

✓ Parents are informed partners in the learning process

家长是学习过程中知情的合作伙伴

✓ External exam readiness is built progressively, not left to the end

外部考试准备度是逐步建立的，而非留到最后

For questions about implementation, contact: Connor Scanlan, Curriculum Developer

如有实施问题，请联系：Connor Scanlan，课程开发人员

END OF DOCUMENT 文件结束

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